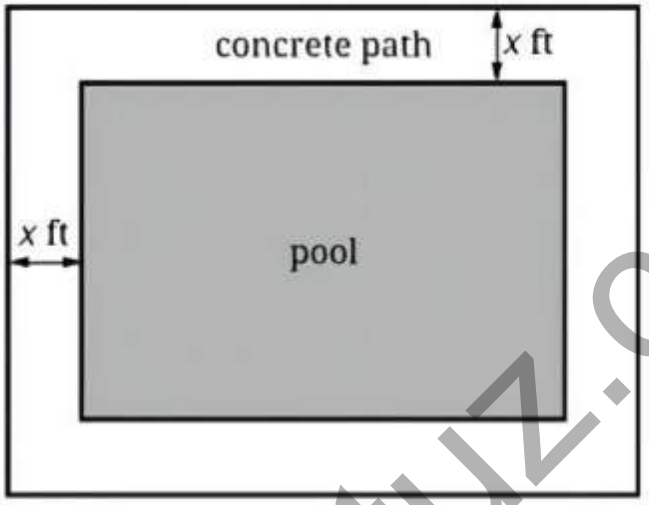
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16 Mark for Review



Note: Figure not drawn to scale.

The figure shows a rectangular pool surrounded by a concrete path that is x feet (ft) wide on all sides. The pool is 21 ft long and 13 ft wide. The area of the concrete path is 240 ft^2. What is the value of x?

3

6

20

40

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22 Mark for Review

$$r^2 + qr = 8r - 97$$

In the given equation, q is an integer constant. The given equation has no real solutio
What is the largest possible value of q?



Answer Preview:

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19 Mark for Review

$$f(x) = 3x^2 + 48x + 193$$

The function g is defined by $g(x) = f(x + 7)$. What is the minimum value of $g(x)$?

(A) -15

(B) -8

(C) 1

(D) 8

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13 Mark for Review

$$f(x) = (x - a)(x - b)$$

The function f is defined by the given equation, where a and b are integer constants. If $f(17) > 0$, $f(20) < 0$, and $f(23) > 0$, what is one possible value of $a + b$?

Answer Preview:

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$$24x^2 - (12a + 2b)x + ab = 0$$

In the given equation, a and b are positive constants. The sum of the solutions to the given equation is $k(6a + b)$, where k is a constant. What is the value of k?

Answer Preview:

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20 Mark for Review

$$kx^2 - 16x = 24x^2 - 8$$

In the given equation, k is an integer constant. If the equation has two distinct real solutions, what is the greatest possible value of k?

Answer Preview: 31

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The quadratic equation $(2x - 64)^2 = a$, where a is a constant, has exactly one real solution. What is the value of a?

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$$x - a = (x - a)(x - 24)$$

Which of the following are solutions to the given equation, where a is a constant and $a > 25$?

I. a
II. 24
III. 25

(A) I and II only

(B) I and III only

(C) II and III only

(D) I, II, and III

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$$r^2 + qr = 8r - 87$$

In the given equation, q is an integer constant. The given equation has no real solutions. What is the largest possible value of q?

Enter your answer (e.g., 5.566, -5.566, 2/3, -2/3)

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18 Mark for Review

The quadratic function g models the depth, in meters, below the surface of the water of a Weddell seal t minutes after the seal entered the water during a dive. The function estimates that the seal reached its maximum depth of 382.2 meters 7 minutes after it entered the water and then reached the surface of the water 14 minutes after it entered the water. Based on the function, what was the estimated depth, to the nearest meter, of the seal 10 minutes after it entered the water?

Enter your answer (e.g., 5.566, -5.566, 2/3, -2/3)

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13 Mark for Review

$$f(x) = -0.0038400x^2 + 15.236x - 15,103$$

The model shown gives the predicted average Arctic sea ice area $f(x)$, in millions of square kilometers, for September of each year x, where x represents a year and $1979 \leq x \leq 2012$. Based on the model, what is the positive difference, in millions of square kilometers, between the predicted average Arctic sea ice area for September of the year 1991 and the predicted average Arctic sea ice area for September of the year 1992? (Round your answer to the nearest thousandth.)

Enter your answer (e.g., 5.566, -5.566, 2/3, -2/3)

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3 Mark for Review

The solutions to $x^2 + 10x + 19 = 0$ are r and s, where $r < s$. The solutions to $x^2 + 10x + 1 = 0$ are t and u, where $t < u$. The solutions to $x^2 + 20x + c = 0$, where c is a constant, are $r + u$ and $s + t$. What is the value of c?

Enter your answer (e.g., 5.566, -5.566, 2/3, -2/3)

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16 Mark for Review

$$f(x) = (x - a)(x - b)$$

The function f is defined by the given equation, where a and b are integer constants. If $f(32) > 0$, $f(35) < 0$, and $f(38) > 0$, which of the following could be the value of $a + b$?

(A) 35

(B) 37

(C) 67

(D) 69

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$$(x-k)^2=(k-4a)(x-k)$$

In the given equation, a and k are constants, where $k>4a$. The sum of the solutions to the equation is $3k+37$. What is the value of a?

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$$y = 4x^2 - bx - 5$$

Which of the following equations is equivalent to the given equation, where b is a positive constant?

A. $y = 4\left(x - \frac{b}{8}\right)^2 - 5 - \frac{b^2}{16}$

B. $y = 4\left(x + \frac{b}{8}\right)^2 - 5$

C. $y = 4\left(x - \frac{b}{8}\right)^2 - 5$

D. $y = 4\left(x + \frac{b}{8}\right)^2 - 5 - \frac{b^2}{16}$

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19 Mark for Review

$$\frac{1}{24}x^2 + \left(s - \frac{1}{24}t\right)x - st = 0$$

In the given equation, s and t are positive constants. The product of the solutions to the given equation is $-2kst$, where k is a constant. What is the value of k?

Enter your answer (e.g., 5.566, -5.566, 2/3, -2/3)

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23 Mark for Review

$$nx^2 - 16x = 26x^2 - 8$$

In the given equation, n is an integer constant. If the equation has two distinct real solutions, what is the greatest possible value of n?

Enter your answer (e.g., 5.566, -5.566, 2/3, -2/3)

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24 Mark for Review

A quadratic function gives the estimated length of daylight $d(t)$, in hours, in a certain city t months after March 1, where $0 \leq t \leq 7$. According to the function, the estimated length of daylight is 12.66 hours 6 months after March 1 and the maximum estimated length of daylight is 13.91 hours 3.5 months after March 1. Based on this function, what is the estimated length of daylight, in hours, on March 1?

Enter your answer (e.g., 5.566, -5.566, 2/3, -2/3)

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